Decentralized Bio-Inspired Multi-Agent/Swarm Coordination and Emergency Protocols for Electric Vehicles

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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

The fast development of the Electric Vehicle (EV) usage has changed the landscape of the city traffic. Though this change to less polluting transportation is a promising one, it has developed additional problems in terms of safe compatibility and smart coordination of cars. Current navigation systems continue to utilize slow, centralized updates when an accident in one or more major intersections is obstructing traffic or when a sudden thunderstorm causes roads to get slippery, which leaves the drivers stuck in traffic jams or ignorant of the hazards on the road. To overcome these obstacles, we suggest a two-fold strategy based on bio-inspired multi-agent algorithms such as ACO (Ant Colony Optimization), PSO (Particle Swarm Optimization) and Bee Colony Optimization to assist electric vehicles in coordination in real-time. This also has the capability of detecting hazards, congestion avoidance as well as adaptive traffic signal synchronization. More crucially, we are creating a universal i.e protocol, a language that any electric vehicle i.e BMW, Nissan Leaf or Tesla can speak to others in times of emergencies. The suggested system will be tested in normal and emergency conditions.

Keywords: Electric Vehicles (EVs), Bio-Inspired Multi-Agent Systems, Emergency Protocols, Swarm intelligence, V2X Communication, Decentralized Coordination, Traffic Optimization, Ant Colony Optimization, Bee Colony Optimization, Particle Swarm Optimization, Urban Mobility.

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Chapter 1

Introduction

1.1 Background

The emergence of electric vehicles is becoming the norm in urban transportation. This change is not simply the replacement of gasoline with electricity. It is transforming completely the way we perceive vehicles. Electric cars are fundamentally computers on wheels, loaded with state of the art sensors. The difference between them and regular fuel-powered cars is that they can process information in real-time. This equips them with features that are absent in normal cars. Nevertheless, the modern traffic management systems are similar to how they were decades ago. They still rely on centralized systems that tend to be slow to such an extent that they are not effective. As an example, when we are stuck in traffic jams because an accident or any other emergency situation has taken place, the information that we require is usually received too late. It begins as a minor inconvenience but later on it becomes a big safety concern.

Moreover, this issue becomes even more problematic when we can take into consideration the manner in which various car manufacturers manage their communication. BMW cars have one "language", Tesla have another, and Nissan have another. It is similar to emergency responders who are unable to communicate with one another in times of crisis. This language barrier demonstrates that, although individual cars are getting smarter, they cannot collaborate with each other.

Bioinspired (swarm) coordination systems have the potential to provide a way out as they replicate the same decentralized intelligence as in nature. We think about the way in which ants seek the shortest way to sources of food devoid of the guidance of the queen or even how bee colonies make decisions as a group regarding the place to construct their hives. These biological networks exhibit three characteristics that can be used in EV networks. Their attributes include scalability with distributed control, robustness with redundancy, and flexibility with local decision-making.

The potential implication goes beyond speculations. Bio-inspired coordination can help reduce the response time in the emergency and achieve up to 20 per cent urban transportation congestion reduction. These protocols would still be in a position to offer transportation services when normal systems fail on natural disasters. Most importantly, this study leads to truly intelligent transportation networks that can act with adaptive and resilient behavior.

1.2 Rationale of the Study or Motivation

The main rationale of the study is the crucial gap in the current utilization of electric vehicles (EVs) as a part of the urban mobility system. These vehicles, even with their state-of-the-art sensing, communication and computation abilities, remains aggressively reliant on fixed, centralized infrastructures, particularly when they encounter stressful emergency events. The current systems fail to provide full support to real- time, cooperative behaviour of EVs that constrain their possibilities in dynamic traffic conditions.
The need to have transport systems that are adaptive and remain operational even when pressured by external factors has risen owing to the increased occurrence of natural disasters, traffic congestion, and breakages of the infrastructure. In addition to that, we can also learn how change is handled by nature and how living systems respond to pressure. Through such principles, it is possible to have a decentralized, bio-inspired framework that can assist in the flexible and reliable co-ordination in uncertain scenarios. This research works to improve transportation by using the full potential of electric vehicles. We can make the roads safer while also helping emergency vehicles move faster.

1.3 Problem Statement

Due to the increasing number of Electric Vehicles (EVs), it has become more difficult to manage modern traffic systems. Despite having many environmental and operational benefits, the integration of EVs into urban traffic raises two critical and unresolved challenges.

1.3.1 Lack of Decentralized Real-Time Coordination

Nowadays, most of the existing EVs act as independent entities. These vehicles still rely on age-old centralized traffic data and basic navigation systems. Though these existing systems work fine in typical everyday scenarios, they struggle a lot with the increased EV densities. Moreover, during any sudden changes in the road these vehicles cannot always act rapidly to the changes. This reflects their unreliability in a dynamic environment. As a result, drivers may experience wrong turns, slow updates, or delayed reactions to hazards.
Furthermore, many new generation EVs come equipped with advanced sensors and data-sharing tools, but those features are rarely utilized to their full potential. It may seem that these electric vehicles could exchange warnings and live traffic updates about nearby problems to reduce risks but in reality most systems lack direct communication. This happens due to interoperability issues among vehicles from different manufacturing companies. Due to this, it becomes harder to reduce congestion or alert drivers to potential risks.
All these limitations of the existing systems become a bigger issue during emergency situations like blocked roads, accidents and sudden weather changes. Those scenarios require fast response and decision making ability which is absent in present day centralized systems.

1.3.2 Lack of a Standardized Emergency Communication Protocol

One of the major challenges in the present day EV sector is the absence of a universal communication system during emergencies. As each of the electric vehicles from different

manufacturers often operate on their respective platforms, they cannot share important information apart from their own brand. For instance, during a natural calamity, one EV from a particular company cannot send alerts or route updates to another EV from a different company even when it is crucial for safety. In addition, lack of common systems which would allow the emergency electric vehicles to interlink and communicate with normal electric vehicles. This implies that the responders are sometimes unable to request passing vehicles to pull over and move aside so that time is not wasted when it is needed most. There is also increased difficulty in handling evacuations by city officials when other kinds of EVs are present on the road.
This communicational barrier is not only limited to emergency responses. The effects of these EVs will be more felt in the future as the cities strive to go smarter. They have a gigantic potential in not just transportation of people but also provision of power in times of emergencies. However, when these systems fail to cooperate, it will be hard to make that idea a reality.

1.4 Objective

This study explores two interrelated solutions

- **Decentralized Coordination:** The application of bio-inspired algorithms such as ACO, PSO and BCO can improve the autonomous behaviors of EVs. Which will ultimately help in adaptive traffic signal cooperation, congestion aware routing and hazard response.
- **Emergency Interoperability:** Developing a standard communication protocol which will support cross-vendor coordination in emergency situations.

The Primary Objectives of this research paper are:

1.4.1 Implementing Swarm Intelligence for EV Optimization

In order to help electric vehicles work together, many nature inspired methods like Particle Swarm Optimization (PCO), Bee Colony Optimization (BCO) and Ant Colony Optimization (ACO) will be used in this research. These algorithms will guide vehicles to choose better routes, share road problems such as blocks, holes, wreckage etc. by collecting information from their sensors. This system will also improve traffic conditions by managing traffic lights in real time. This happens by adjusting the traffic lights based on how busy the roads are. All of these will be carried out without the need of a central system. Thus, traffic flow will be smoother and more responsive.

1.4.2 Developing a Common Emergency Communication Protocol

The main aim here is to create a lightweight communication protocol which will allow electric vehicles from different brands to share important information during emergencies. This will make it easier for ambulances, fire trucks, police and other emergency vehicles to move through traffic seamlessly by communicating with nearby EVs to clear the route. The system will have a common message format so that vehicles of different brands can

exchange information among themselves . As a result we can achieve more efficient communication without any compatibility issues.

1.4.3 Simulating and Testing the System in Realistic Scenarios

The whole system is tested using tools like SUMO and OMNeT++. The simulations include both regular traffic and emergency circumstances.The system is evaluated based on its responsiveness, travel time, emergency response time, coordination with other vehicles as well as its capacity to deal with unexpected problems.

1.4.4 Evaluating Scalability and Practical Use

The proposed new system will be evaluated to see how it compares with traditional traffic control methods. The comparison will focus on how well the system adapts to change, how safe it is, how well it handles faults and how easily the system can be added to current EV networks and infrastructure.

1.5 Methodology in Brief

This research is based on simulation driven experiments that helps us to test intelligent routing strategies for Electric Vehicles (EVs) for both normal and emergency traffic conditions. The research is performed using the SUMO microscopic traffic simulator, which is controlled externally using TraCI (Traffic Control Interface), so that the algorithmic-based dynamic adaptation of routes can be performed in real time.
The methodology is divided into three basic stages. Initially, a baseline traffic scenario is created with real-world road network data in which vehicles follow static, shortest path routing without any adaptive intelligence. This baseline is used as a reference for comparison against performance. In the second stage, models for existing routing intelligence are implemented and evaluated. These are Ant Colony Optimization (ACO), Dijkstra's shortest path algorithm, A star search and an existing hybrid of ACO-PSO combined. Each of those get integrated into SUMO via external python controllers. These controllers dynamically monitor traffic conditions such as congestion levels, travel time, waiting time and reroute vehicles using algorithm specific decision logic while ensuring route continuity constraint imposed by the simulator. In the final stage, the research moves towards a hybrid and emergency aware routing framework where routing decisions are triggered by events (e.g. congestion spike, route blockage, presence of emergency vehicles, etc.).
This framework focuses mainly on communication driven routing , which allows vehicles to change routes based on shared traffic conditions and local observations. Also a single metrics logging framework is used to guarantee a fair and uniform comparison across all algorithms. In the end, we can say that the overall methodology provides reproducibility, algorithmic fairness and a realistic traffic behavior. As a result it provides a strong foundation for analyzing intelligent EV routing in dynamic and emergency circumstances.

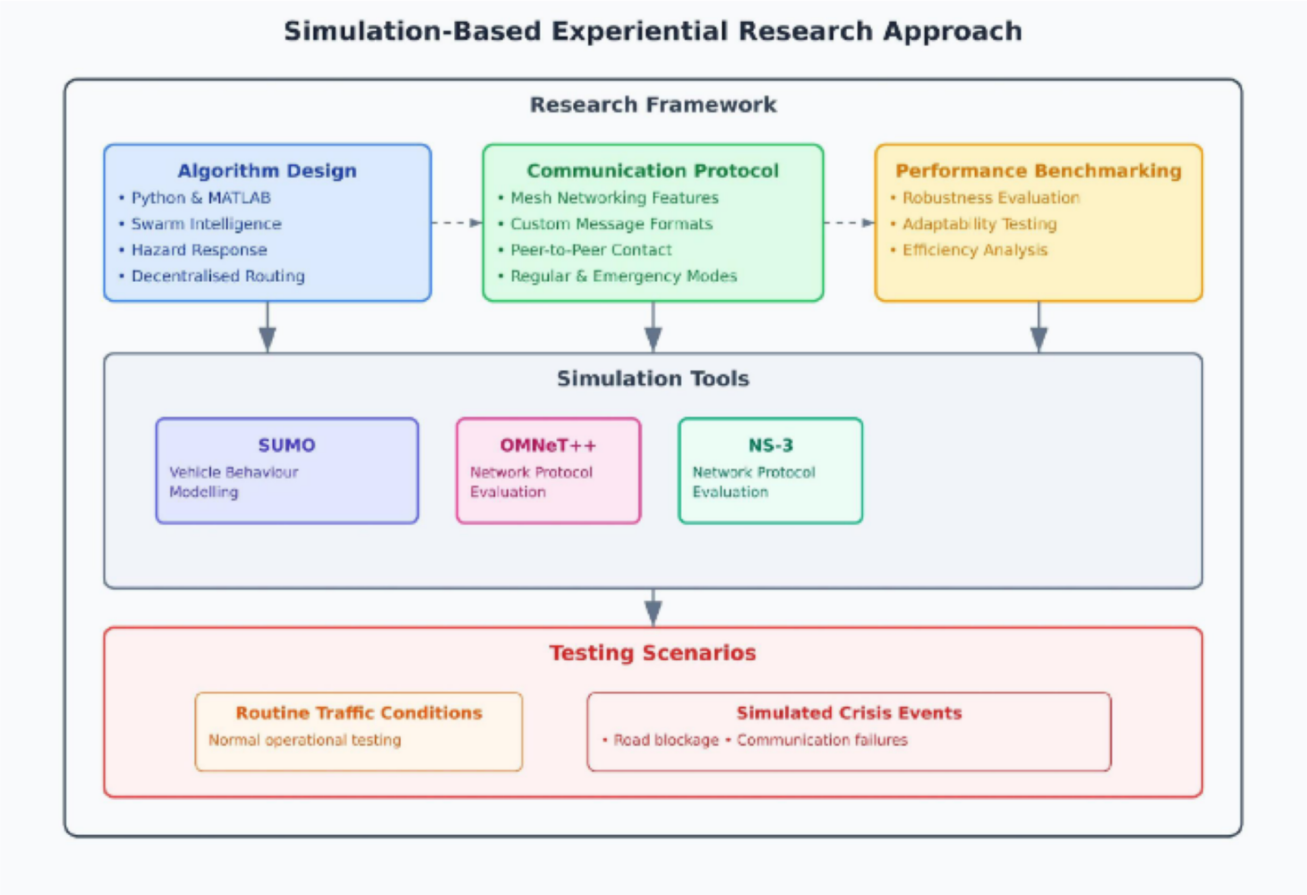


Figure 1.1: Research Methodology Diagram

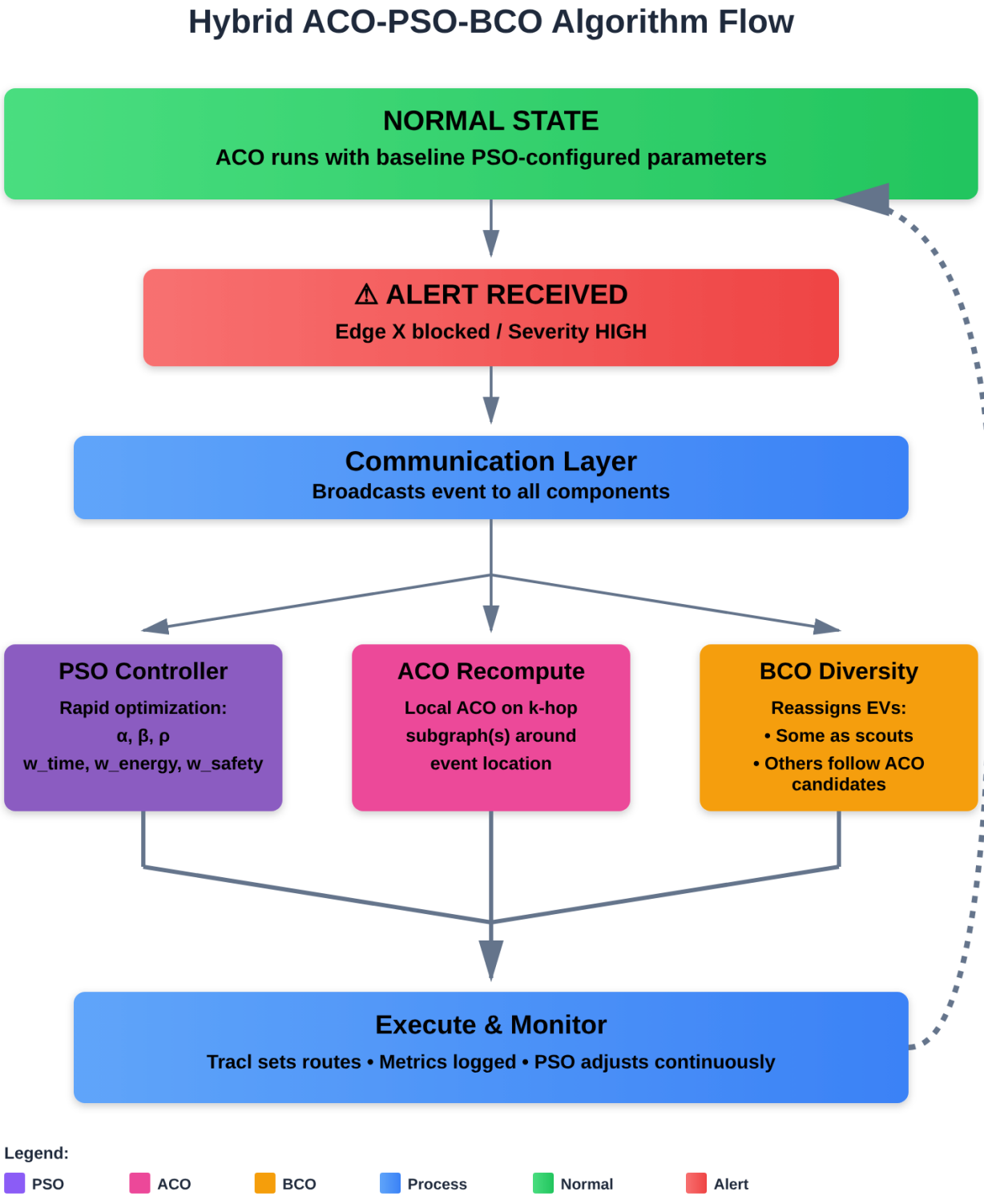


Figure 1.2: Experimental Methodology Diagram

1.5.1 Workflow plan

The research follows a well structured workflow to test intelligent routing algorithms in electric vehicles both in normal and emergency traffic scenarios. It follows a simulation based approach. In the initial step, a real urban traffic network is downloaded from OpenStreetMap (OSM) in the .osm format. This file is later converted into (network.net.xml) format using the netconvert tool which is SUMO compatible. This tool retrieves the lane structure, road topology, traffic regulations and road junctions from the actual map. The simulation is conducted using two different methods. In one method, the road network is populated by vehicles with random trips across the map using the SUMO utility (randomTrips.py). In another method, real vehicle trips were integrated from a real life dataset. All these vehicle trips are stored in an xml file which contains the start and destination node pair of the trips of each vehicle. All these trip definitions of each vehicle are transformed into a detailed route that are valid and continuous through the network. These routes are stored in another file of (.rou.xml) format.

The entire simulation environment is configured using a single .sumocfg file. This file defines the simulation duration and execution parameters along with the integration of all the necessary inputs. The entire project directory is divided into several logical folders, such as algorithms, controller, metrics, and results having separate role in the system functionality. The algorithms folder contains all the used routing algorithms like Dijkstra, A* search, Ant Colony Optimization (ACO) and an existing hybrid of ACO and PSO. The controller folder contains algorithm specific controller scripts that interact with the simulation through TraCI (Traffic Control Interface). These scripts dynamically modify vehicle routes and monitor traffic conditions at simulation runtime.

Initially, the vehicles follow normal routing behaviour in the absence of emergencies. But emergency scenarios like fire, accident or road blockages are introduced randomly in the simulation which forces the adaptive rerouting to be triggered based on the active algorithm. TraCI is used along with unified metric logging scripts to gather performance metrics such as travel time, waiting time, distance traveled, and emergency response time. All the algorithms are tested in the same traffic conditions and the data obtained is stored in CSV format. Those results are later used for comparison and performance analysis.

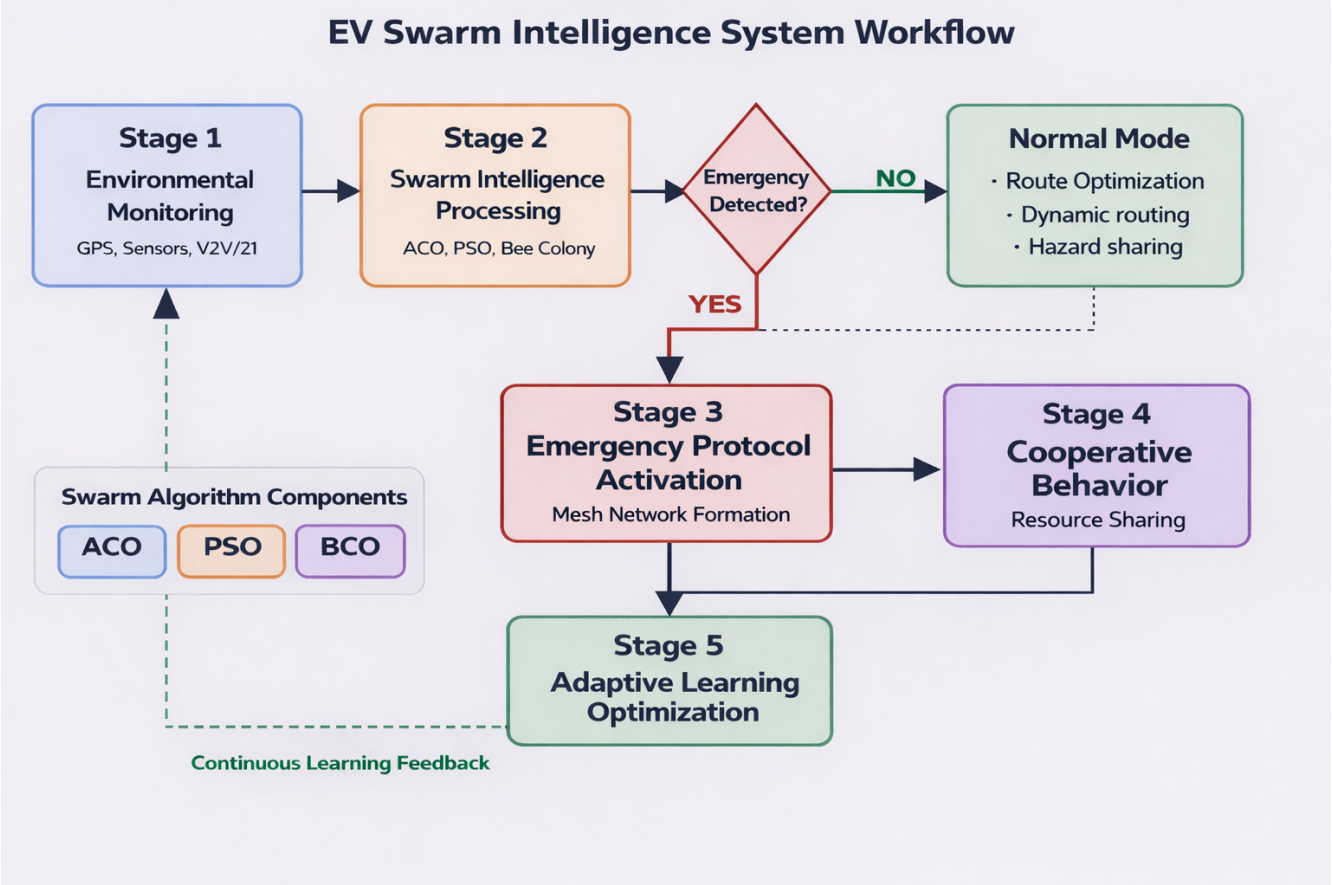


Figure 1.3: Electrical Vehicle Swarm Intelligence System Workflow

1.6 Scopes and Challenges

1.6.1 What's Included in the Research

- Simulation-based validation of 3 swarm algorithms (ACO, PSO, Bee Colony) for EV coordination
- Urban traffic scenarios (normal and emergency conditions)
- Vehicle densities: approximately 8000 EVs
- Emergency interoperability protocol design
- Performance comparison against centralized systems

1.6.2 What's Limited

- Simulation only; no real-world EV testing due to cost and safety constraints
- Urban focus with limited highway and rural testing

1.6.3 Major Challenges

While the proposed methodology provides a structured approach to address EV coordination challenges, several technical and implementation obstacles are expected. Early identification of these issues enables proactive mitigation strategies and realistic project planning.

Technical Challenges

- Real-time performance: Algorithms must execute within 100 ms in vehicles
- Scalability: Simulating 6000+ vehicles within computational limits
- Communication realism: Modeling packet loss and interference in V2X networks

Validation Challenges

- Limited baselines: Few comparable swarm-based EV systems exist
- Generalization: Ensuring results extend beyond specific test scenarios
- Interdisciplinary complexity: Balancing computer science algorithms with automotive engineering

Chapter 2

Literature Review

2.1 Preliminaries

Unlike traditional vehicles, EVs can serve as intelligent and decentralized coordination systems because they have wireless communication, sensing and processing abilities. The basis for integrating multi-agent systems and bio-inspired algorithms is swarm-based intelligence. Most types of swarm intelligence are found in literature by observing social behavior in nature, such as the foraging of ants, the flocking of birds, or the coordination seen in bee hives. These systems work with decentralized control and a few simple rules that explain the local actions of individual agents, leading to complex overall behavior. The three main swarm-based methods currently being explored in transportation are Ant Colony Optimization, Particle Swarm Optimization, and Bee Colony Optimization.

- **ACO:** Inspired by the pheromone-laying behavior of ants, is widely used for shortest-path problems in dynamic environments [1]. It operates through the probabilistic selection of paths based on historical usage and environmental feedback.
- **PSO:** Based on the flocking of birds or schooling of fish, treats every agent as a particle that adjusts its trajectory based on personal experience and that of neighboring agents [2]. This makes it suitable for clustering and load-balancing in vehicle systems.
- **BCO:** Inspired by the food foraging and recruitment behavior in bee colonies, helps to distribute agents to jointly explore, examine, and exploit resources or pathways [3].

2.2 Swarm Intelligence in EV Systems

There are many benefits of integrating bio-inspired swarm intelligence into EV control systems. The work in [4] shows real-world examples of integrating swarm intelligence into EV control systems. As a result energy consumption is reduced. There are other additional advantages of bio-inspired algorithms. For instance, Martín-García et al. show how anti-pheromone-based foraging algorithms can improve multi-agent collaboration in urban navigation and reduce exploration times [5]. Their study developed a method that copies biological trail evaporation (anti-pheromones) for preventing agents/vehicles from returning to areas they have already visited, so that cars can effectively roam cities. It can be

also used for blockage detection. The system was tested in simulations using both random maps and actual cycling route data from four European cities. It demonstrated improved scalability and adaptability as the number of agents increased.
On the other hand, Seongin Na et al. presented a virtual pheromone-based collision avoidance technique for autonomous vehicle swarms that is enabled by deep reinforcement learning [6]. Reinforcement learning helped the agents to dynamically optimize their navigation courses, lowering the rate of inter-vehicle collisions, especially in congested urban environments. Similarly, Zhai et al. advanced this concept by modeling coordinated obstacle avoidance in automated vehicle swarms in connected environments [7]. Their model included path negotiation and vehicle-to-vehicle (V2V) data sharing, which assisted in a predictive approach for cooperative vehicle behavior. This is the reason the model may anticipate the intentions of other cars, leading to less abrupt maneuvers and more stable navigation patterns in difficult road situations.
Rahman et al. introduced a swarm algorithm based energy allocation framework where autonomous agents can dynamically respond to energy demand fluctuations using probabilistic rule-based behavior [8]. The agents communicated through stigmergic signaling, an indirect coordination mechanism inspired by social insects to balance grid loads and vehicle energy consumption. The framework demonstrated better convergence rates and energy distribution stability in comparative simulations against classical deterministic models.
Kraiem et al. took a complementary approach by integrating Particle Swarm Optimization (PSO) with a photovoltaic (PV) system mounted on electric vehicles [9]. During high solar exposure conditions their energy control strategy dynamically adjusted the power harvested from PV panels to maximize autonomy. Overall, these works show how by combining swarm intelligence and bio-inspired algorithms can lead to more efficient and resilient urban mobility systems. Decentralizing decision-making among autonomous agents can also help to increase more scalability, adaptive capacity and resilience in the next generation of electric transportation systems.

2.3 Hybrid Methods

Using Hybrid methods can help us in energy efficiency and management. For example Afzal et al., who proposed a hybrid method combining CSO with Open Ended Learning (OEL) to optimize State of Charge (SoC) estimation in batteries of EV [10]. By combining swarm intelligence with an adaptive learning framework, their model could react to dynamic changes in environmental conditions and energy usage patterns more precisely. When compared to conventional SoC estimation algorithms, their experiments revealed a significant increase in prediction accuracy and convergence speed, which indicates that the hybrid method is more suitable for practical implementation in autonomous electric vehicles.
Bagwe contributed to this domain by exploring hybrid energy management strategies tailored for HEVs (hybrid electric vehicles) through a modeling approach that incorporates both rule-based logic and cost-optimization algorithms [11]. The authors performed simulations and developed a dynamic model of a parallel hybrid drivetrain, in order to determine the fuel economy and energy efficiency under various load conditions. Asadi developed this thread even more by predictive energy management based on traffic signal and congestion preview data [12]. Using vehicle-to-x communication to predict stop lights and traffic con-

gestion, the suggested approach allowed proactive decisions including energy redistribution or regenerative braking before arriving at congested intersections. Simulated case studies indicated that predictive control could save significant fuel and reduce idle time when compared to reactive energy management solutions.
On the contrary, Ai et al. showed how well Radial Basis Function (RBF) neural networks combined with fuzzy logic could capture the nonlinear dynamics of urban traffic flow which is often impacted by random events like accidents or signal failures [13]. Their hybrid model allowed for both accurate short-term forecasting and adaptive recalibration, making it especially suited for real-time traffic prediction and congestion mitigation in smart urban settings. Compared to conventional statistical methods, their system showed improved robustness and lower prediction errors across multiple datasets. The Studies collectively show that hybrid methods improve efficiency, adaptability and real-world performance in electric vehicles in a significant way. The result provides vehicles with energy management, predictive abilities and environmentally sustainable factors.

2.4 Emergency Situation Handling

While swarm and hybrid AI systems can improve energy and traffic management in general scenarios, emergency situations can create additional challenges. Coordination becomes difficult by infrastructure breakdown, poor communication and the need for prioritizing emergency vehicles (e.g., ambulances or fire trucks). Dang et al. highlights that bad weather conditions can severely affect traffic sign and hazard recognition in EV systems, which underlings the need for enhanced perception models with robust feature extraction to maintain situational awareness [14].
A solution can be a dynamic signalization model. Siddiqi et al. developed a dynamic signalization model for emergency vehicle navigation in smart cities, applying AI-driven traffic control to ensure real-time responsiveness [15]. Their system used predictive modeling to anticipate the movement of emergency vehicles and adjust traffic signals in advance. The model applied multiple variables such as traffic density, signal cycles, and emergency vehicle speed to generate adaptive signal plans that minimize delays and clear routes efficiently. Simulation results confirmed improved response times and reduced congestion spillover compared to static signalization systems.
On the other hand, interoperability challenges in EV infrastructure were addressed by Karpenko et al. emphasizing that interoperability in EV charging and parking systems, that can seamlessly exchange data across heterogeneous platforms requires semantic-level compatibility in both hardware or protocol-level interoperability [16], which can also guides us to create a common Emergency Communication Protocol. Another approach can be short-range wireless communication. Aligning to our goal, the United States Department of Transportation emphasized the benefits of short-range wireless V2V communication for sharing location, speed and emergency alerts in real-time [17]. The report concluded that V2V technologies have the potential to drastically reduce collision rates and emergency response times when deployed at scale.
Another way to reduce the challenges Hu et al. proposed a V2X-based emergency material scheduling system capable of real-time resource allocation during disaster response scenarios [18]. Their framework incorporated vehicle-roadside communication, cloud-based route planning, and predictive resource deployment. Simulation-based validation proved its

effectiveness in enhancing the responsiveness of emergency logistics, particularly in high-congestion disaster zones where real-time coordination is critical. A complementary study by Nellore and Hancke proposed a visual-sensor-based traffic signal prioritization system for emergency vehicles [19]. By the help of computer vision and dynamic signal control the system can detect incoming emergency vehicles and modify traffic signals accordingly. It addressed the common drawback of fixed-cycle traffic lights, which often fails to adapt to the requirements of the road in real time. In simulations that resemble busy urban intersections, their method greatly decreased delay and increased throughput for emergency responders.
Decentralized control techniques also appear to be promising in emergency scenarios. By integrating a mobile application with GSM-based communication, Siddiqi et al. created a smart signalization framework that enables emergency vehicle operators to actively control traffic lights [15]. Their model empowers drivers to pre-activate green signals for intersections ahead using an SMS-triggered interface. Through a field-tested prototype deployed in Turkey, the authors demonstrated that the system significantly reduced emergency response time along with improving overall intersection safety. The study concluded that decentralized control paired with low-power GSM modules offers a scalable and affordable solution to the problems related to existing urban transportation in emergency situations. To prioritize emergency vehicles, the recent research [20] introduces a framework which combines both V2V/V2I communication and adaptive traffic lights. Compared to fixed-cycle systems, the emergency response times is reduced by 40% by this system as it uses GPS, IoT, and YOLOv8 to dynamically adjust signals. This highlights the promise of decentralized systems in emergency situations, despite ongoing scalability limitations. Collectively, these studies demonstrate that these solutions can reduce response times, prevent congestion spillover and ensure the prioritization of emergency services. So by integrating these models and creating an interoperable protocol, we can ensure safety and benefits that can change the future.

2.5 Summary of Key Findings

To begin with, Bio-inspired algorithms such as ACO, PSO and BCO can help to decentralize coordination among electric vehicles, which allows the EV to operate collaboratively. These bio-inspired algorithms improve dynamic navigation, scalability, collision avoidance and energy management by mimicking natural systems like ant foraging, bird flocking and bee communication. Adding to that, Combining bio-inspired algorithms with hybrid methods (reinforcement learning, fuzzy logic and radial basis function neural networks) has improved predictive capabilities, energy efficiency and traffic forecasting. In emergency scenarios AI-driven signal control, V2X/V2V communication and visual detection systems reduces response time, prevents congestion spillover and ensures priority routing for emergency vehicles. To ensure safety on both normal and emergency conditions in EV, resilient sensing systems (for bad weather), semantic interoperability in infrastructure and collaborative vehicle behavior are essential. To conclude, the studies highlight the importance of interoperability, bio-inspired algorithms and hybrid methods along with real-time decision-making , adaptive behavior in creating resilient and efficient electric vehicle ecosystems.

Table 2.1: Summary of Reviewed Literature

No.	Objective	Framework	Models Used	Datasets Used	Results
1	Framework for embodied cognition and robotics	Subsumption Architecture	Conceptual design	–	No numerical performance; conceptual framework
2	Optimization of nonlinear functions using PSO	PSO	PSO (bird flocking-based)	Standard optimization benchmarks	Demonstrated effectiveness on benchmark problems
3	ABC algorithm inspired by bee foraging	Artificial Bee Colony (ABC)	Standard optimization benchmarks	Benchmark functions	Superior global optima finding ability
4	Reduce EV steering system current via swarm tuning	PID-PSO and PID-ACO	PID-PSO, PID-ACO	Simulation data from EV systems	Current reduction; MSE converged in 15 iterations
5	Anti-pheromone EV navigation via swarm algorithm	Anti-pheromone swarm algorithm	Anti-pheromone swarm algorithm	Synthetic maps; 3 EV prototypes	Reduced exploration time; feasible, robust design
6	Decentralized collision avoidance via DRL + pheromones	PhERS, DDPG, HLER, Noisy Networks	Turtlebot3 + Gazebo traffic simulations	Turtlebot3 + Gazebo	100% success; faster training, smoother trajectories
7	Obstacle avoidance in connected AV swarms	A2C DRL, Cooperative lane-change model	A2C DRL	3-lane freeway sim, varied obstacles	Speed +15–34%, flow +10–31% vs. baseline
8	Swarm-based PHEV charging station optimization	PSO, APSO, GSA, PSOGSA	PSO, APSO, GSA, PSOGSA	Simulated charging scenarios	Best fitness, convergence by PSOGSA
9	PSO-based MPPT for EV solar charging	MPPT (P&O vs. PSO1/PSO2)	PSO	Simulated PV with irradiation/motion	0.5–1% more power than P&O; faster tracking
10	SOC estimation using CSO and OEL	CSO-OEL hybrid + feature engineering	CSO-OEL	Simulated + real battery data	Improved estimation, convergence, battery health
11	Adaptive EMS for hybrid EVs (ARBS)	ARBS + comparative models	ARBS	UDDS cycle + dynamometer tests	SoC stability, fuel reduction, real-time feasibility

Continued on next page

Table 2.1 – *Continued from previous page*

No.	Objective	Framework	Models Used	Datasets Used	Results
12	MPC-based fuel saving using signal preview	MPC, DP, macroscopic models	VISSIM + PSAT	SUV scenarios	47% fuel savings, smoother dynamics
13	Hybrid fuzzy-RBF for urban speed forecasting	RBF NN + fuzzy logic/-clustering	RBF NN	Traffic speed data with conditions	MAPE 6.4%, better than baseline models
14	Weather-robust traffic sign detection	YOLOv5s+SE, YOLOv5s+C3GC	YOLOv5s	Weather-variant traffic datasets	Accuracy: up to 79.2% with C3GC model
15	GSM-based ERV signal override system	Arduino + GSM SIM808 + mobile control	Mobile App	Field urban test deployments	Reliable activation; faster ERV clearance (qualitative)
16	IoT data exchange framework for EV smart parking	O-MI/O-DF, mobile IoT app	O-MI/O-DF	Smart parking and EV charge data	Functional interoperability in smart city deployment
17	V2V emergency communication framework	DSRC + BSM protocols	DSRC	Connected vehicle field scenarios	Framework validated (no exact accuracy reported)
18	Emergency material scheduling using V2X	SEMS + fairness model	SEMS	Disaster response simulation data	Improved scheduling via scenario coefficients
19	Vision-based traffic management for ERVs	PE-MAC, visual sensors + WSN	PE-MAC	Intersection and sensor event data	Less delay than IEEE 802.11p
20	Adaptive signal + EV comms for smart cities	YOLOv8, GPS, MQTT, IoT	YOLOv8 + GPS	GPS + live traffic camera feeds	Emergency response time reduced by 40%

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

This thesis proposes the design and testing of an emergency-conscious hybrid swarm routing framework of electric vehicles (EVs). The system is decentralized, and during dynamic emergency conditions, it operates on adaptive route optimization. The implementation and evaluation of the system are carried out in a microscopic traffic simulation environment (SUMO) using the TraCI (Traffic Control Interface). The adaptive re-routing behavior is triggered by road blockages, unexpected congestion, or a failure of infrastructure. The framework works on simple inter-vehicle and vehicle-to-infrastructure communication so as to distribute information on emergencies.
Some of the metrics that are used in assessing system performance include average travel time, congestion distribution, routing convergence time, and emergency response effectiveness. The final deliverables will be a working prototype of the simulation-based model, experimental outcomes of the simulation under various emergency conditions, and the comparison of the results with the baseline routing models.

3.2 Societal Impact

The proposed study will deal with the paramount problems in the transportation system in cities, specifically under the emergency conditions. The system can help to mitigate the congestion and enhance the overall traffic flow during disruptive situations by allowing vehicles to respond dynamically to emergency events by adjusting routing decisions. This may help in quicker response to emergencies in addition to safe evacuation.
The interoperability between vehicles without reference to centralized traffic management systems is facilitated by the decentralized aspect of the proposed framework. This strategy encourages equitable and collaborative routing conduct amongst heterogeneous vehicles that is especially vital in big urban settings where vehicles are not owned by any particular company, and there are different types of infrastructures.
The study also contributes towards the development of intelligent transportation systems by showing how intelligent transportation systems can be benefited by swarm intelligence methods to offer adaptive, resilient, and scalable routing solutions. The results can be used in future research and development to enhance emergency traffic management and smart city mobility solutions.

3.3 Environmental Impact

The research is carried out in a fully simulated setting, and it does not produce any direct negative effect on the environment during development. The system indirectly serves to enhance efficiency in traffic, through minimization of unnecessary congestions and rerouting delays in cases of emergency.
Better traffic performance with obstacles on the road can shorten idle periods and ease the movement of vehicles that can lead to less energy consumption and emissions during the implementation process. Although environmental optimization is not the main aim of this thesis, the given framework is oriented on sustainable transportation principles, since it helps to coordinate and efficiently route vehicles.
Findings of the research can be used as a base to develop future research on the matter that explicitly addresses the issues of environmental and energy-related considerations in emergency-conscious routing systems.

3.4 Project Management Plan

The project will be developed by a structured and incremental development plan. First step is to understand the knowledge of the fundamental elements of ACO, PSO, and BCO and their application. To enable independent testing and validation, these components are written in a modular manner. Preparations of simulation environments and baseline routing scenarios occur simultaneously.
The next phases are the integration of the algorithms into a hybrid system and emergency-induced routing adaptation. Experimental testings are done in controlled scenarios to examine the behaviour and performance of the system. Progress review is also done within a weekly meeting with the supervisor to make sure that they are working towards the project goals.
Teamwork and updated work are also ensured through a common GitHub repository. Algorithms are handled in a modular design and incremental testing to deal with potential risks, including complexity in implementation and integration issues. The last stage is the result analysis, documentation, and writing the thesis.

Chapter 4

Design Process and Methodology

4.1 Methodology Overview and Experimental Scope

This study is based on the experimental simulation to compare and analyze the electric vehicle navigation behavior to various routing algorithms in the normal and emergency traffic scenarios. The primary aim is to know how the traditional shortest path algorithms and the bio inspired swarm based algorithms behave when there is a sudden disturbance like accidents, congestion, fire or road blockages. The methodology of the research has a systematic plan. Initially, a real life urban road network is simulated by using SUMO which is a microscopic traffic simulator. Then vehicles are integrated into the road map with a predefined given start and end point pair. Various algorithms such as deterministic shortest path algorithms are used to compute the routing decisions of vehicles along with some adaptive swarm-based approaches. Many emergency scenarios are introduced in the simulation at runtime by dynamically altering the road conditions in random times. This forced the routing algorithms to respond to network topology and traffic flow changes. The same experimental setup is maintained for the simulation of all the algorithms to ensure reproducibility and fairness. All the performance metrics like travel time, waiting time, emergency response time, congestion levels are logged throughout the entire simulation which allows a direct evaluation and comparison of algorithm efficiency, adaptability and robustness under normal and emergency traffic conditions.
Combining the controlled vehicle simulation along with systematic collection of performance metrics, this methodology provides a reliable analysis of the routing performance, strengths and limitations of each of the algorithms in dynamic real life traffic environments.

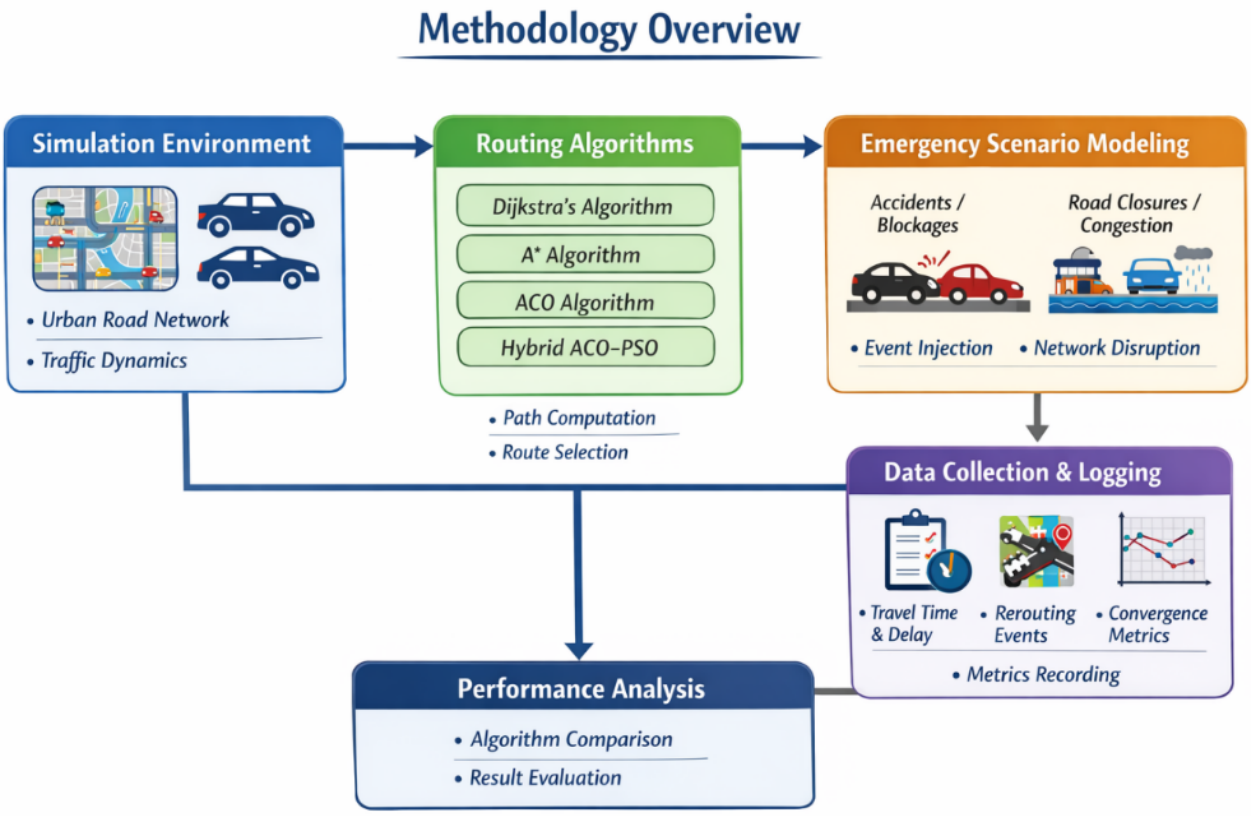


Figure 4.1: Methodology Overview

4.2 Simulation Environment and Traffic Modeling

The simulation environment is built using a microscopic traffic simulator. Simulation of Urban Mobility (SUMO) platform is being used which offers a detailed control of vehicle behavior, road network dynamics and traffic lights. The road network is represented as a directed graph where road segments are represented as weighted edges and intersections as nodes. Edge weights represent the properties of the travel cost like distance, level of congestion or travel time. The traffic demand of the vehicles are created in two approaches: first one is conducted by generating random trips and the second one is conducted based on predefined route and trip files. The simulator provides realistic acceleration, deceleration and car following models that the vehicles follow. This very configuration is suitable to model both the behavior of individual vehicles and the emergent traffic patterns at the network scale. Emergency scenarios are introduced in the simulation at runtime by dynamically altering the road conditions in random times. This involves blocking certain edges, lowering the capacity of roads, or increasing the level of congestion on certain routes. These modifications mimic the interference with the real world like accidents or emergency road closures. The introduction of sudden emergency events forces the vehicles to reconsider their paths depending on the capacities of the routing algorithm in operation. Except the routing logic of the algorithms all the other aspects of the simulation environment are kept unchanged. This ensures that the observed differences of the performance are due to algorithmic behavior rather than variations of other aspects of the environment.

4.3 Data Collection

The source of data of the present research is the NYC Taxi and Limousine Commission (TLC) Trip Record Data,from which we took the Yellow Taxi trip records of March 2025. We chose this dataset because it has a detailed representation of taxi activities. As it is under the official NYC TLC validation and can be practically used in traffic modeling.
The raw data consists of about 4.1 million trip records with 20 attributes each has temporal (pickup and dropoff times), spatial (location IDs), trip (distance, number of passengers) and fare (amount of payment, surcharge) information. The information was obtained at the official NYC TLC site (https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page) and which makes the data authentic and reliable to research-related topics.
Additional information was used such as the NYC Taxi Zone Lookup CSV file with borough classifications , zone descriptions and taxi zone shapefiles ,which afterwards helped us convert location IDs to specific latitude and longitude points.

4.4 Data Transformation

The data transformation method was divided into multiple stages in order to prepare the raw TLC data for traffic simulation in SUMO. This trip filtering is based on the Manhattan taxi zone's specifications. In which, we were able to find the location IDs of every Manhattan borough (a total of 69 zones) by using the taxi zone lookup table on their website. Only trips with pick up (PULocationID) and drop off (DOLocationID) locations inside the specific Manhattan area were included in the data after additional filtering. This reduced

down the dataset to 4.1 million intra-Manhattan trips to about 3.1 million trips.
The GeoPandas was then used to combine filtered trip data with NYC taxi zone Shapefiles in order to perform geographic coordinate mapping. The location ID was mapped to the corresponding geographic coordinates, where the pickup location is assigned to the pickuplat and pickuplon columns and the dropoff location assigned to the dropofflat and dropofflon columns.
After that nearly a total of 3,100,000 filtered trips resulted in the systematic extraction of 9,000 trips with proper proportional representation with respect to time, locations and trip characteristics. This sample size is reasonable because it allows calculating traffic patterns and is statistically sound. After that we converted the 9000 trips into SUMO compatible routes and mapped them to actual road segments in SUMO's Network. We discarded the 2% of trips which were landing on invalid locations. After that we filtered the taxi accessible roads as there were edges for trains, bikes and pedestrians. Another problem was solved by reusing working route pairs as Manhattan's street network was fragmented and the origin destination pairs were not connecting, which dropped the success rate to 6.2%. Lastly , we sorted the trips by time as rerouting needs to be in a chronological order and generated the final XML files.

4.5 Findings of Analysis of the Preprocessed Data

The overall processed data set of Manhattan taxi traffic simulation is 9,000 exemplary trip records that have comprehensive spatial and temporal information to perform effective simulation modeling in SUMO. The data format contains such key attributes as VendorID, pickup and drop off time (tpep pickup datetime, tpep drop off datetime), number of passengers, distance of the trip, rate code, type of payment, breakdown of the fare, geographic coordinates (pickup lat, pickup lon, dropoff lat, dropoff lon). Data quality was ensured during preprocessing where we did verification procedures. So that there were no null values in key coordinate variables, correct temporal analysis (valid format of a timestamp) and logical consistency between pickup-dropoff pairs.The resulting geocoded and clean processed data offers a solid base of simulating realistic Manhattan traffic and allows investigating the spatial patterns of taxi movements, route optimization and modeling the traffic flow within the operational bounds.

4.6 Routing Algorithms Implemented

This research compares four routing techniques from conventional shortest path algorithms and the bio inspired swarm based algorithms. The exact simulation setup is maintained for the simulation of all the algorithms to ensure fairness.

4.6.1 Dijkstra's Algorithm

This algorithm is considered as a classical shortest path routing baseline. It explores the network in a systematic manner and reduces the cumulative edge cost. In this manner it

computes the optimal path from source to destination. After the computation of the optimal path, that particular route remains fixed unless it is explicitly recalculated or modified. Although the Dijkstra algorithm provides optimality in static traffic networks, it lacks efficiency and responsiveness in dynamic real world environments where sudden disruption occurs.
The core mechanism involves repeatedly relaxing edge distances:

```
for neighbor, weight in graph[currentnode]:
    newdist = dist[currentnode] + weight
    if newdist < dist[neighbor]:
        dist[neighbor] = newdist
        parent[neighbor] = currentnode
```

The above code snippet comprises the core mechanism of Dijkstra's algorithm which repeatedly relaxes edge distances in order to compute the shortest path for a vehicle from a source node to all other nodes. This gives a non-adaptive routing policy in the simulation which provides a static and baseline routing strategy.

4.6.2 A^* Search Algorithm

A^* algorithm is an extension of the shortest path search that provides a heuristic function to steer exploration in the direction of the destination in a more efficient manner. Combining actual path cost with an estimated remaining cost, A^* usually decreases the computation time and search space compared to Dijkstra's algorithm. Nevertheless, similar to Dijkstra, its performance in emergency is limited as its efficiency in dynamic emergency conditions depend on the frequency with which replanning is activated.

```
f_score[neighbor] = gscore[neighbor] + heuristic(neighbor,
   destination)
openset.put((f_score[neighbor], neighbor))
```

The shortest path search is improved by A^* algorithm as it combines the actual path cost (g) with a heuristic estimate (h). This helps to select routes much faster in large urban networks compared to Dijkstra's algorithm.

4.6.3 Ant Colony Optimization (ACO)

One of bio bio-inspired algorithms that is used in the simulation is Ant Colony Optimization. It uses a probabilistic approach to construct paths which is based on heuristic information and pheromone trails. The algorithm gradually adapts to traffic conditions when the artificial ants i.e vehicles continuously explore the traffic network and reinforce promising routes over time. This algorithm performs well under dynamic environments compared to the other two due to its iterative and distributed character. Despite this, rapid network changes may result in slower convergence.
The implementation of the decision logic in our controller is as follows:

```
# Calculate probability based on pheromone and distance
prob = (pheromone[u][v] ** alpha) * ((1 / distance[u][v]) ** beta)
```

```
# Probabilistic selection of the next intersection
nextnode = select_node_based_on_probability(prob)
```

ACO copies the foraging behavior of ants by choosing routes based on edge desirability and the intensity of pheromones. As time passes, the efficient paths that are used frequently are more likely to accumulate higher levels of pheromones and thus adaptive route optimization is achieved.

4.6.4 Hybrid ACO–PSO Routing Algorithm

The hybrid ACO-PSO algorithm is a fusion of path construction mechanisms of ACO with rapid parameter modification of the Particle Swarm Optimization (PSO). Here, ACO creates candidate routes and PSO continuously modifies key algorithm parameters in response to the random changes in the traffic conditions due to emergency or sudden changes. The main goal of this hybrid approach is to reduce convergence speed and improve responsiveness by keeping the flexibility of swarm-based routing intact. The integration of PSO into the ACO controller is implemented as a feedback loop within the simulation runtime:

```
# Generate candidate routes using current ACO logic
routes = aco.generate_routes(traffic_state)

# PSO updates alpha and beta based on current performance metrics
aco.alpha, aco.beta = pso.update_parameters(routes,
   performance_metrics)
```

In this case, candidate routes are created by ACO while PSO continuously adjusts important ACO parameters based on the real time traffic and emergency conditions. This feature improves responsiveness and provides better convergence speed during any sort of disturbance.

4.7 Implementation Architecture

The implementation follows a modular and extensible architecture in order to guarantee clarity, reproducibility and fair comparison between routing algorithms. Each routing algorithm is constructed as a separate module with a shared interface, allowing for easy switching between algorithms without affecting the simulation environment.
The traffic simulator communicates with the routing logic via a runtime control interface allowing real-time route updates and network state monitoring. As a result, we do not need to change our metric logging component every time we simulate different algorithms. At each step of the simulation, relevant traffic information is gathered such as the edge travel time, the congestion level and the vehicle position and is then sent to the routing module. Based on this information, the algorithm decides whether or not to re-compute routes and computes new paths, if needed.
A centralized metrics logging component is used to record performance data during the simulation. That includes per-vehicle travel statistics, reroute events and convergence related information of all algorithms. All logged data is stored in structuring formats to enable statistical analysis and visualization later.

This architecture ensures that performance variances are caused by algorithmic behaviors mainly, rather than implementation bias .Our architecture may be expanded and used with other routing techniques in the future.

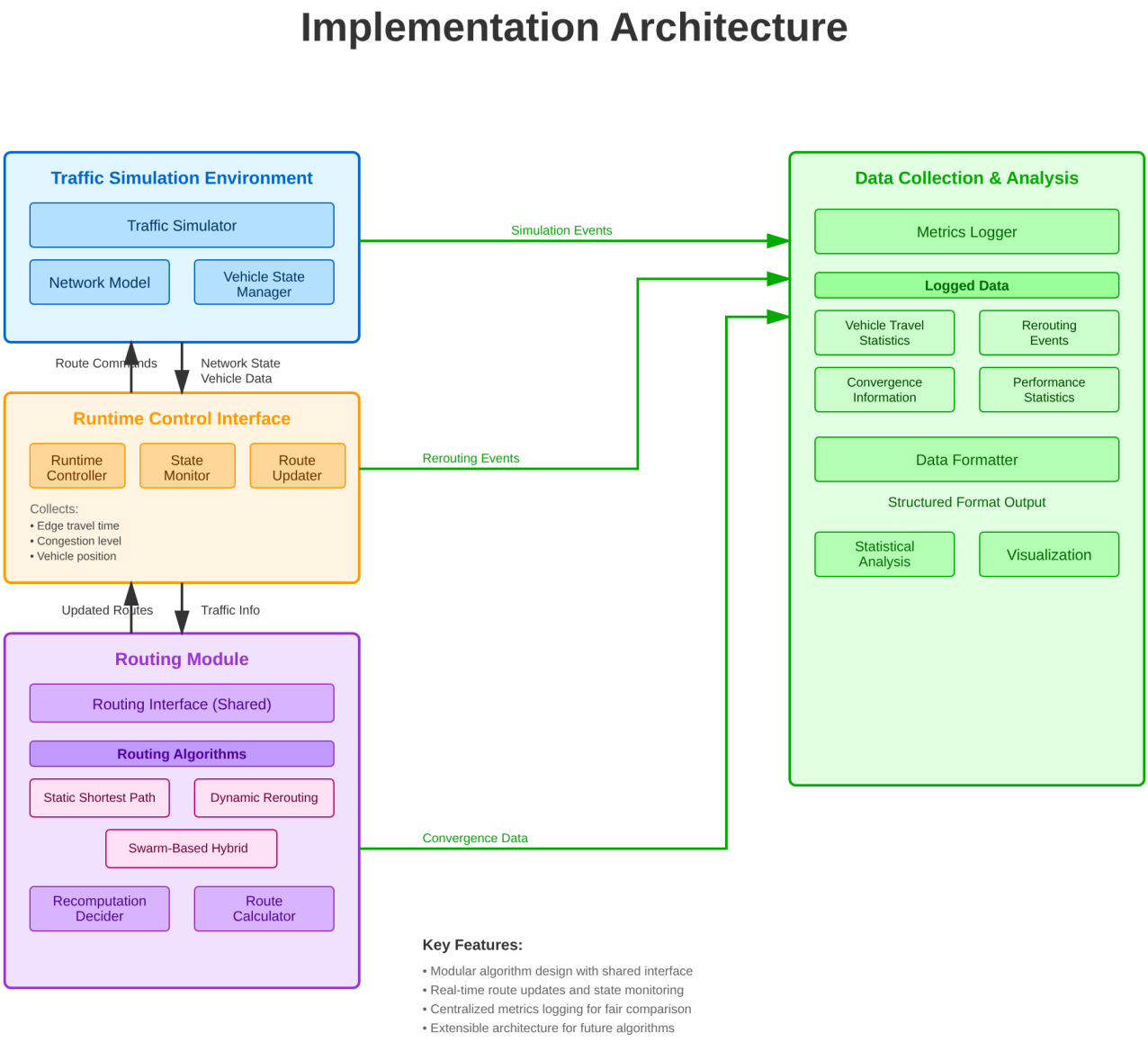


Figure 4.2: Implementation Architecture

Chapter 5

Results And Discussion

Performance Analysis: Emergency-Aware Swarm-Based EV Routing A number of quantitative performance metrics are measured during every simulation run in order to systematically assess the performance of the routing algorithms implemented. These measures are travel time, waiting time, distance covered and congestion which shows the performance of the individual vehicles and also the traffic performance of the whole area. The other indicators like frequency of rerouting and cumulative number of rerouting is applied to determine the flexibility of the algorithm during dynamic and emergency situations. All the data is automatically logged at regular timings through a common data measurement system, and all algorithms are tested under the same traffic and emergency conditions to provide fairness. The resulting data functions as a strong basis on which comparative analysis of performance, responsiveness and stability can be done.

5.1 Performance Evaluation

The three measures that were used to evaluate the framework are accuracy, reliability and efficiency. The measurement of accuracy was done by average travel time (275.4s vs 287.9s baseline for random trips and 269.2s vs 287.9s for predefined trips), waiting time (2,012.6s vs 2,099.2s) and distance optimization (1,383.3m vs 1,443.2m). The measure of reliability was through simulation stability with 6,737 vehicles with uniform rerouting rates (62.5Experiment used microscopic traffic simulation on SUMO on March 2025 data of NYC TLC Yellow Taxi trip data. Out of the total 4.1M records, 3.1M intra-Manhattan trips were selected based on taxi zone lookup data, which created 9,000 representative trips. The anomalies that were injected consisted of accidents, road blocks and infrastructure failures. The performance indicators indicated that ACO-PSO experienced a 4.1% waiting time improvement, 4.2% distance improvement and 4.3% travelling time improvement relative to the non-adaptive routing along the Manhattan road network.

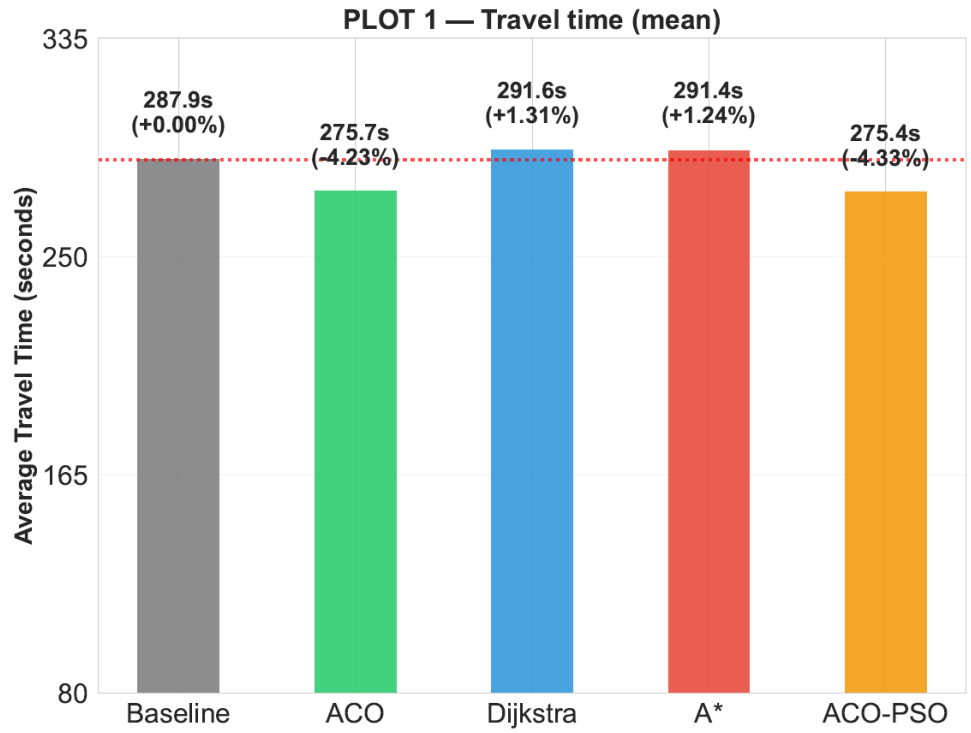


Figure 5.1: Travel Time for Random Trips

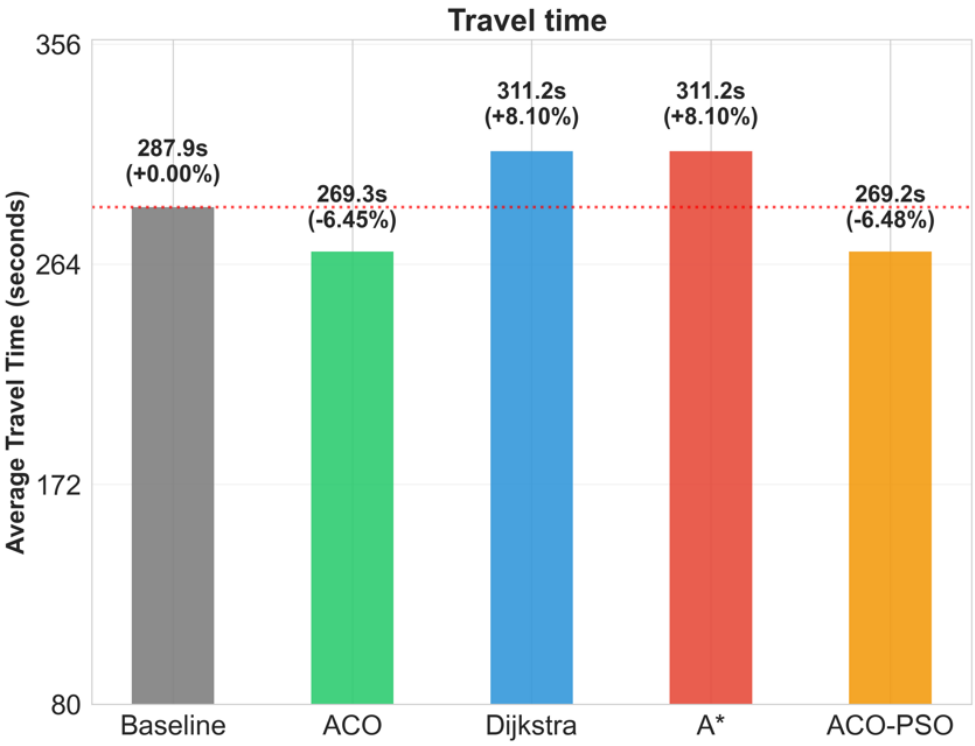


Figure 5.2: Travel Time for Predefined Trips

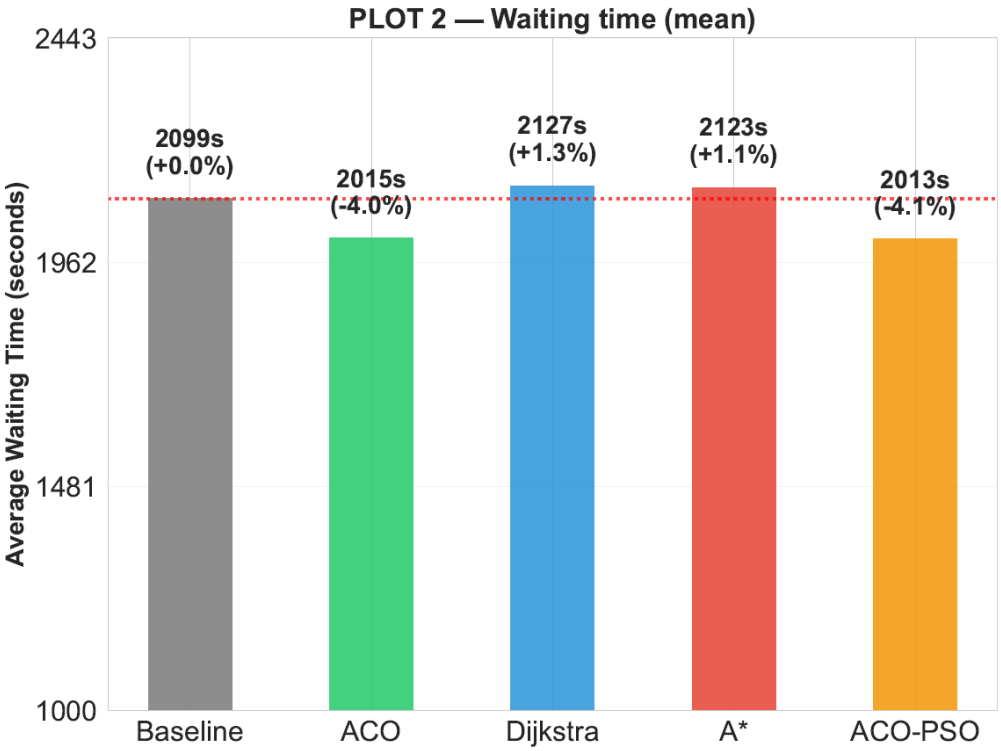


Figure 5.3: Waiting Time for Random Trips

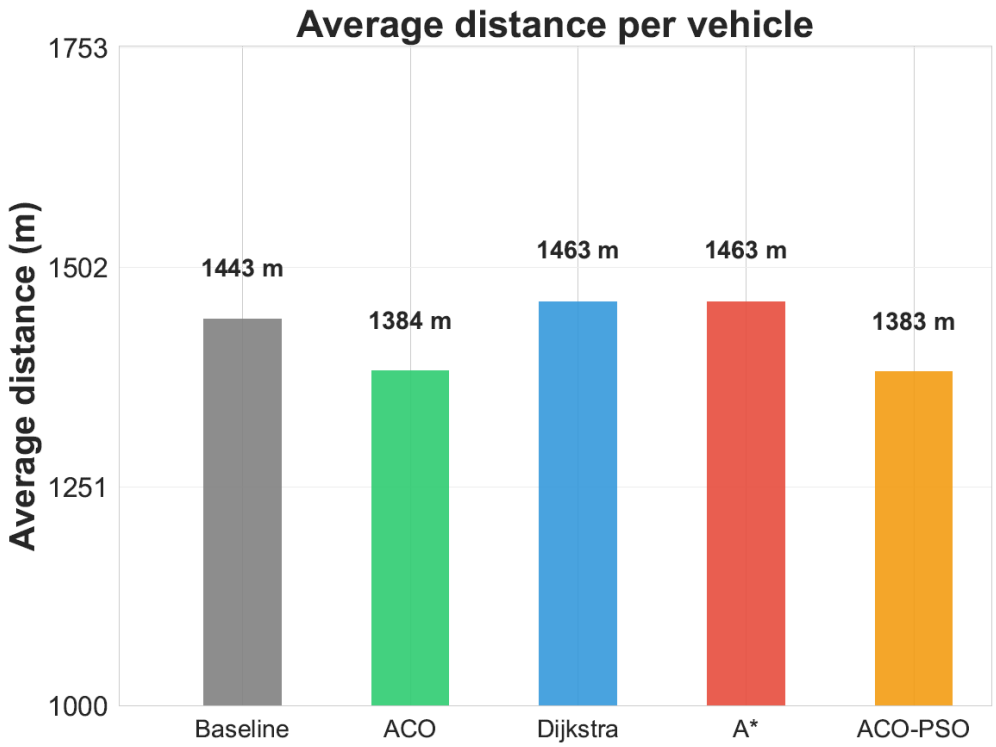


Figure 5.4: Average Distance per Vehicles for Random Trips

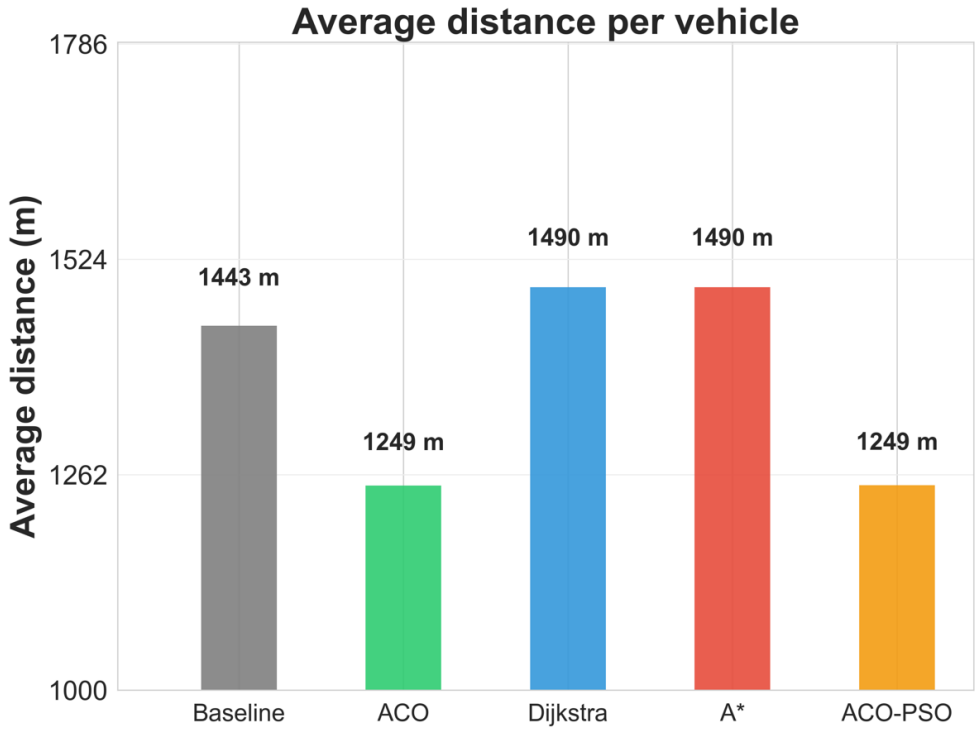


Figure 5.5: Average Distance per Vehicle for Predefined Trips

5.2 Analysis of Design Solutions

Five routing methods were used. Namely Baseline(SUMO default), Dijkstra, A, ACO and ACO-PSO hybrid. The best results were attained by ACO-PSO that uses a combination of pheromone-based path discovery and particle swarm velocity optimization. It yielded the best metrics in all the categories. The cost of implementation was average . It needed distributed computation infrastructure and no custom hardware. Feasibility is high given the SUMO compatibility and scalable architecture. Strengths include dynamic rerouting capability (62.5vehicles were rerouted) as well as emergency response flexibility and a decentralized decision-making that minimized single-point failures of the design. Weaknesses include increased computational cost (1.8 reroutes/vehicle), more variation in standard deviation (1,594.38s vs 1,563.93s baseline) and potential communication latency in real deployments The traditional algorithms (Dijkstra, A)showed negligible improvement, whereas swarm-based models performed significantly better than the baseline, which confirmed the bio-inspired optimization in the dynamic traffic situation.

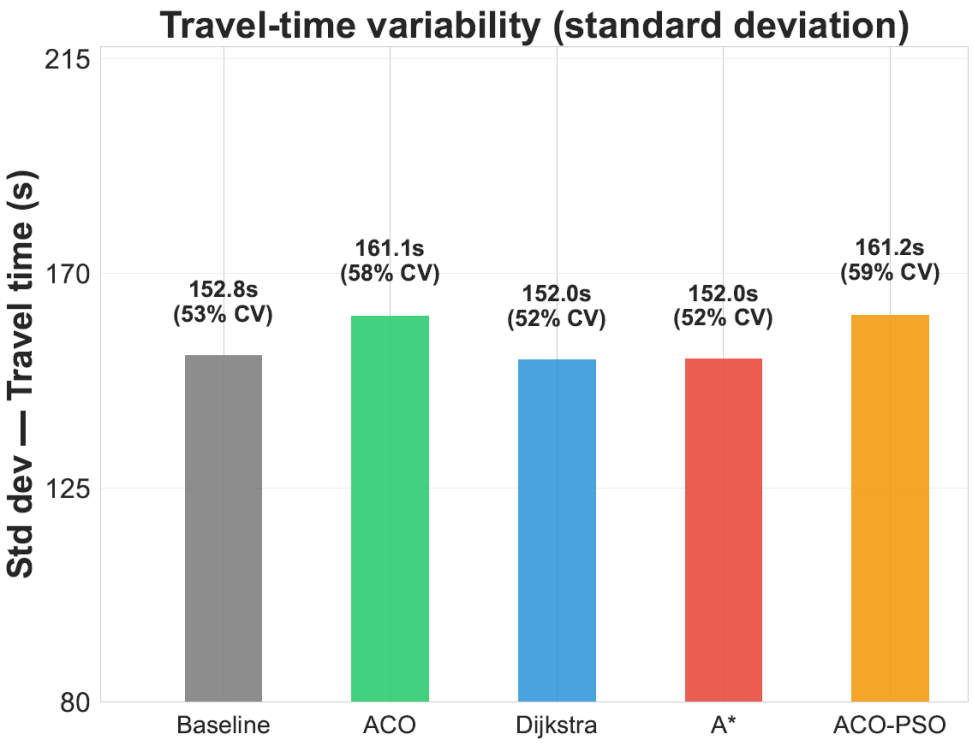


Figure 5.6: Standard Deviation Comparison for Random Trips

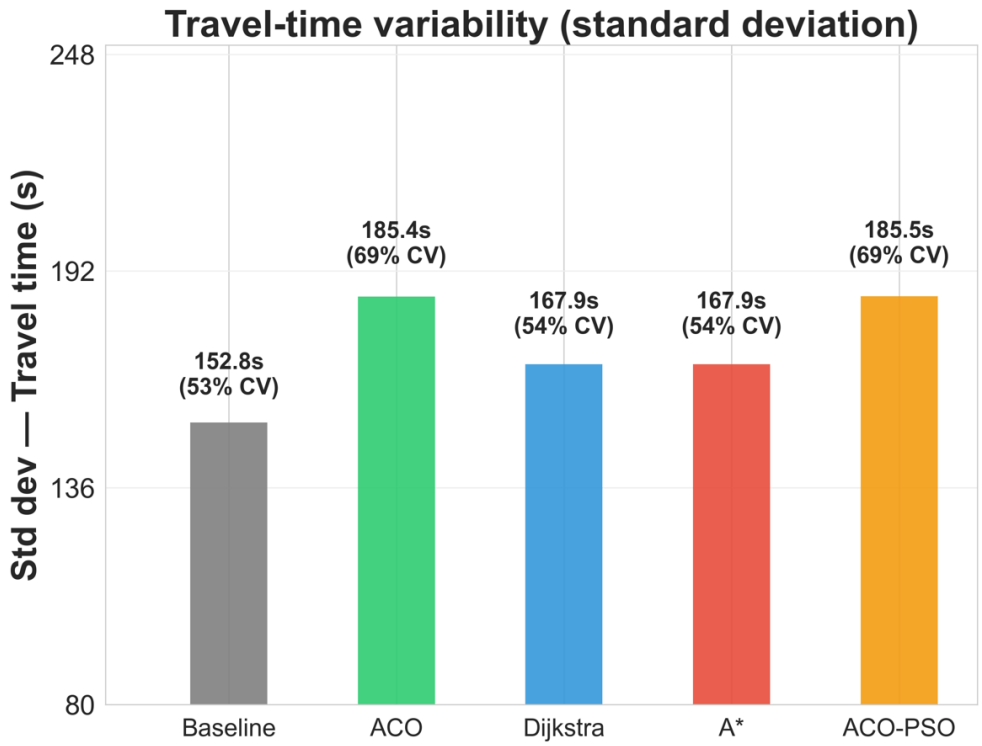


Figure 5.7: Standard Deviation Comparison for Predefined Trips

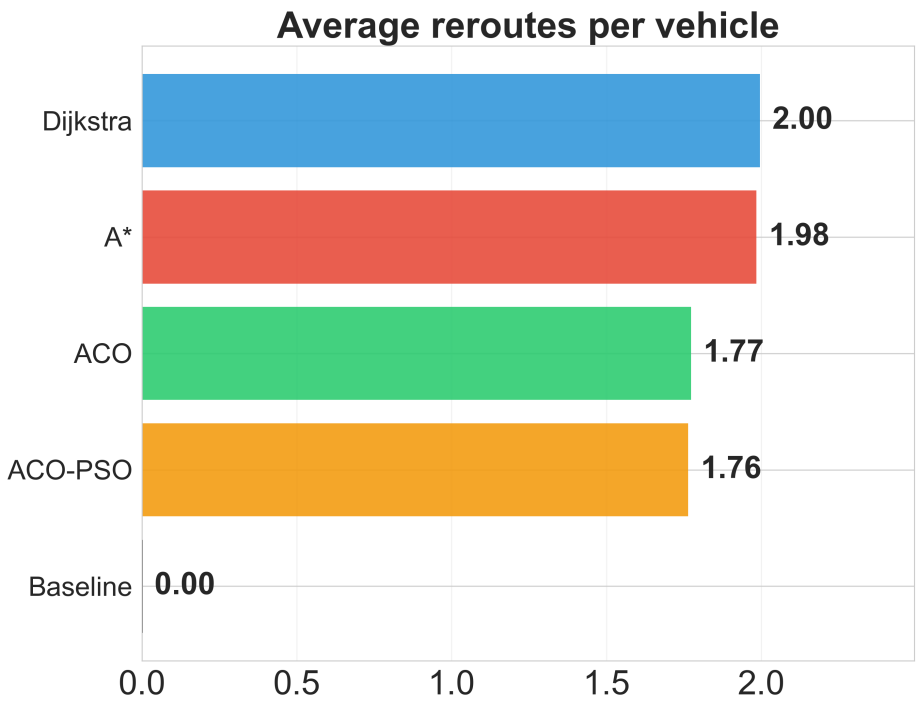


Figure 5.8: Average Reroute comparison for Random Trips

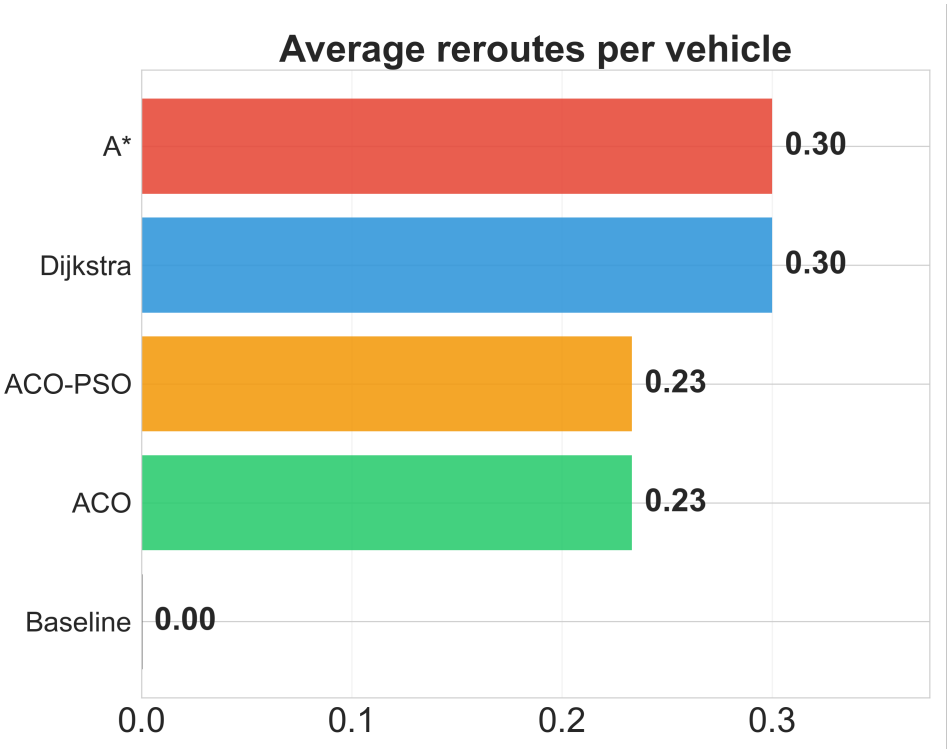


Figure 5.9: Average Reroute comparison for Predefined Trips

5.3 Comparisons and Relationships

ACO-PSO performed better than all the other algorithms. It was 4.1% better than baseline, 5.4% better than Dijkstra and 5.3% better than A* in waiting time. Swarm algorithms were dynamic (as compared to deterministic pathfinding A) with an 62.5% rerouting efficiency. In comparison to the already existing centralized EV routing systems, our decentralized system removed bottlenecks in the servers and allowed emergency response in real time. Conventional algorithms demonstrated 64% rerouting with increased reroute counts (2.0 vs 1.8), which implied that they were inefficient at selecting paths. The hybrid character of ACO-PSO hybrid was a combination of exploration of ACO and convergence speed of PSO, which resulted in 0.3s faster travel time compared to pure ACO. Distance optimization had indicated that ACO-PSO made a 4.2 percent cut in comparison to 1.4 percent (A*/Dijkstra) and was better at multi-objective optimization. The emergency-awareness capability provided adaptability which is absent in baseline approaches.

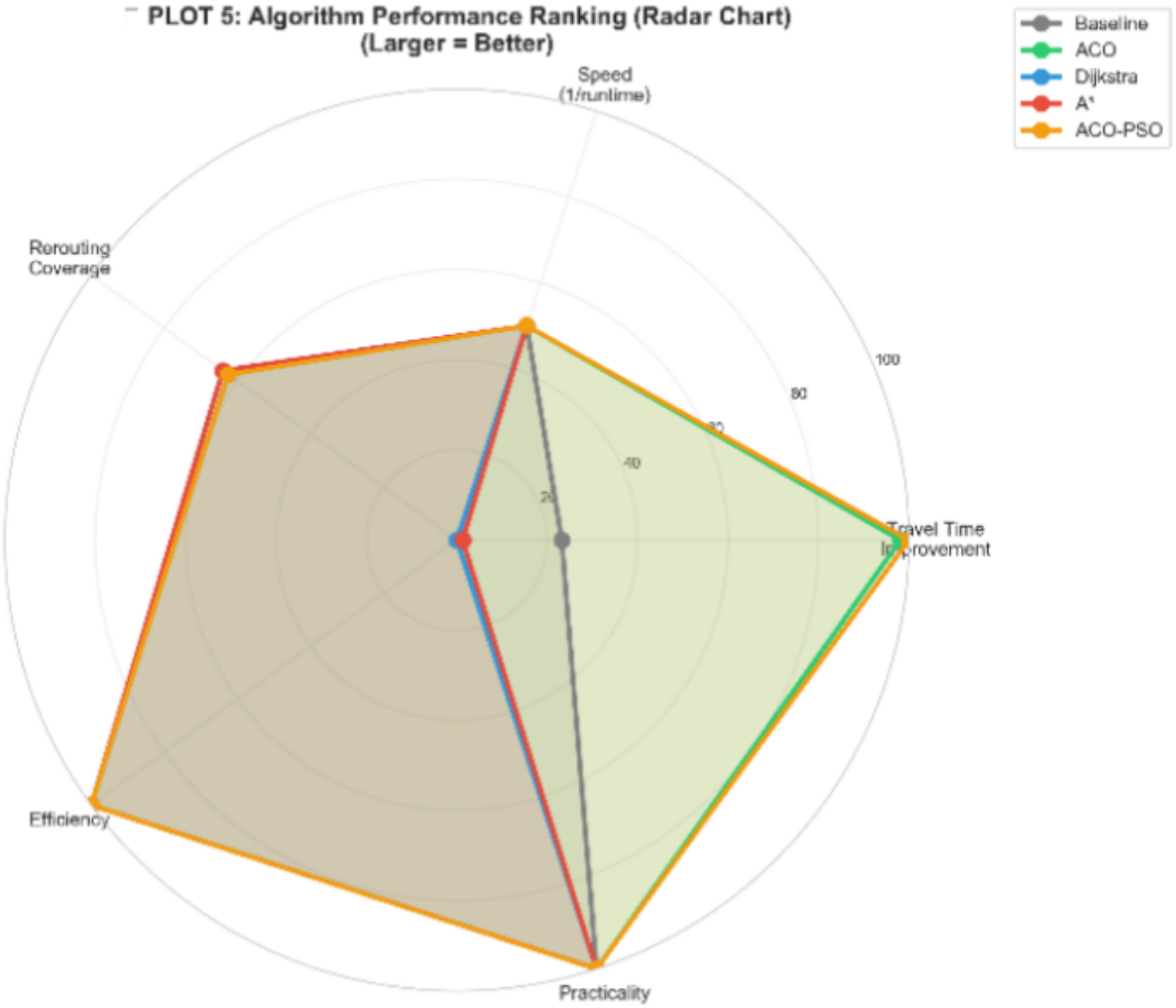


Figure 5.10: Algorithm Performance Comparison

5.4 Discussion

The results confirm the use of swarm algorithm based routing to handle dynamic traffic. This is because ACO-PSO manages fewer failures compared to its competitors because it operates under decentralized decision-making and adapting rerouting as a way to deal with centralized system constraints. Higher variance of the swarm algorithms represent an adaptive behavior but they lead to reliability issues at extreme conditions(edge cases). The 62.5% rerouting rate implies even-handed exploration-exploitation trade-offs. Limitations include single region testing (Manhattan), testing of homogeneous vehicle types and also simulated anomaly events (accidents, roadblocks, infrastructure failures). Actual application can be hampered by communication lag and GPS error as well as compliance issues on the part of drivers. The work in the future needs to be validated in the wide variety of urban settings, include a mixed fleet of vehicles and be tested in the situation of real emergency to be fully evaluated.

Chapter 6

Conclusion and Future Work

At the present stage, the baseline routing and intelligent algorithms such as the Ant Colony Optimization (ACO) algorithm, the Dijkstra algorithm, A* search, and the current ACO-PSO hybrid model were introduced and experimented under the same traffic conditions. The performance measures like travel time, waiting time, distance travelled and response to congestion were measured through TraCI-based monitoring. The comparative analysis shows the effective performance of intelligent routing methods compared to non-adaptive baseline routing, especially in dynamically varying traffic conditions.

To carry on with work in the future, the simulation will be prolonged in three significant directions. Firstly, several emergency vehicles including ambulances, fire trucks, police, and relief vehicles will be incorporated, and each of them will have its own goals and priority constraints. Second, a novel three-way hybrid routing algorithm combining ACO, PSO, and Bee Colony Optimization (BCO) will be developed to further enhance adaptability and decision accuracy. Finally, inter-vehicular message passing will be introduced using LTE-based communication, enabling vehicles to exchange traffic and emergency information in JSON format. These enhancements aim to create a fully communication-driven, emergency-aware intelligent traffic ecosystem.

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